

Measuring Validator and Sandboxed-Execution Coverage for LLM-Generated NetOps Artifacts: A Paired Confirmatory Study

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Abstract—We preregistered and executed a paired confirmatory experiment (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e) that measured the marginal detection benefit of adding sandboxed emulation to a deterministic validator for LLM-generated intent→configuration artifacts. Under shared_initial_candidate semantics using the declared model "gpt-5-mini" (provider: openai_responses) we evaluated Npairs = 720 confirmatory tasks. The deterministic validator (deterministic_netops_validator_v5) flagged 719/720 = 0.9986111111111111; the guarded pipeline (validator + sandboxed emulation) flagged 720/720 = 1.0. Paired difference (guarded - baseline) = 0.001388888888888884; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI (10000 resamples, seed = 1729) = [0.0, 0.004166666666666667]. Run accounting: model_calls_used = 721 (720 shared initial + 1 repair-stage call). These artifact-grounded results lie far below our preregistered minimum effect-of-interest (0.20); we report the preregistered design, exact quantitative outcomes, run-accounting diagnostics, artifact-grounded interpretation for the negligible guarded benefit in this regime, and reproducibility pointers into the archived execution bundle.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration and NetOps workflows. Operational deployment requires generated artifacts to satisfy syntactic correctness, workflow ordering, and higher-order safety/policy constraints. A common mitigation architecture is generate→validate→(repair)→execute, sometimes augmented by sandboxed emulation to reveal runtime-only failures that static validators miss. Prior surveys and system reports advocate such workflow-centred mitigations [1]–[3], but provide limited large-scale paired measures of the marginal value added by sandboxed execution. Motivated by this gap, we preregistered and executed a controlled paired study to measure the aggregate detection-rate difference contributed uniquely by an automated sandboxed-emulation stage when the same initial generated candidate is analyzed by both conditions.

Research question and contributions. Our preregistered research question was: for synthetic intent→configuration tasks under shared_initial_candidate semantics, what is the paired detection-rate difference (guarded - baseline) between (A) a deterministic validator-only pipeline and (B) the same deterministic validator followed by automated sandboxed emulation? Contributions: (1) a machine-readable preregistration and faithful autonomous execution; (2) an artifact-grounded

paired analysis on Npairs = 720 with complete accounting; (3) an artifact-grounded interpretation explaining a near-zero marginal effect; (4) concise reproducibility pointers into the archived bundle and prescriptive boundary conditions where guarded stages are likely to matter.

II. RELATED WORK

Workflow-centred mitigations and verification tools. Recent survey and architectural reports document the practical trend of combining LLM generation with deterministic validators, emulators, and repair loops to raise operational assurance for NetOps artifacts [1]. These works emphasize that correctness in intent→configuration pipelines is multidimensional: syntactic parsing, command-ordering/workflow constraints, invariants (e.g., keep-one-uplink-up), and dynamic runtime properties may each require different V&V primitives. Tool-focused case studies highlight the evolution and practical limits of static verifiers: mature configuration-analysis systems have progressively expanded rule-sets and topology-aware checks but still exhibit gaps for runtime-dependent behaviors and stateful sequences [2]. That evolution shows why researchers and operators often layer an emulator or sandbox after static validation: emulation can exercise temporal behaviors, device state transitions, and interactions that are difficult to capture exhaustively as static rules.

Empirical and prototype evidence. Prototype systems that integrate generation with verification and execution-harnesses report heterogeneous gains: in some task regimes a verifier alone catches most dangerous errors, while in others emulation uncovers subtle runtime failures or environment-specific side effects that static rules miss [1], [3]. The practical implication drawn across this literature is that the marginal utility of a guarded (validator+emulation) pipeline depends on (a) validator coverage relative to the task distribution, (b) the fidelity of the emulator, and (c) the distributional properties of the input intents (e.g., constrained, parser-validated intents versus noisy natural-language requests). These prior empirical and system-level observations motivated a paired experimental design: by reusing the identical initial generated candidate across conditions (shared_initial_candidate semantics) we isolate only downstream guarded-stage effects (emulation observations or permitted repair invocations) rather than conflating them with generation variability.

Gaps and evaluation needs. While surveys and system reports argue for generate→validate→repair and for emulator-assisted validation as best practice for high-assurance NetOps, they do not provide large-scale, preregistered paired estimates of the marginal detection benefit under controlled generation semantics. The literature also documents important open questions that our study was designed to probe: how much additional coverage does sandboxed emulation provide when a strong deterministic validator already targets the same workflow patterns, and which task regimes (adversarial inputs, cross-vendor CLI heterogeneity, or stateful multi-step procedures) are most likely to reveal nontrivial guarded-stage benefits? By situating our paired confirmatory design amid these prior findings, we provide an artifact-grounded quantitative datapoint addressing one narrow slice of the broader evaluation agenda advocated in [1]–[3].

III. METHODOLOGY

Preregistration and experimental design. We preregistered a confirmatory paired-binary design (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e). Primary estimand: paired_success_rate_difference_guarded_minus_baseline aggregated across confirmatory samples. Planned confirmatory sample: Npairs = 720 (planned episodes = 1440). Minimum effect-of-interest for power planning = 0.20 (20 percentage points). The preregistration specified inclusion/exclusion rules, missingness handling, multiplicity handling for exploratory secondary tests, and deterministic contamination controls.

Task bank, vendors, and inclusion. Confirmatory items were synthetic intent→configuration tasks from intent_configuration_repair_v1 covering three abstracted vendor-CLI families and three topology templates (leaf-spine, 3-node service chain, triangle). Inclusion required vendor-specific parser acceptance; the preregistered missingness policy permitted up to two deterministic reseeds for parser failures; irrecoverable parser failures would be deterministically excluded and replaced. In execution there were zero irrecoverable parser exclusions: complete_pair_count = 720 (see analysis/missingness_summary.csv).

Model configuration and sampling semantics. Declared/requested model: "gpt-5-mini" (provider field: openai_responses). The archived model configuration records max_output_tokens = 2000 and temperature = null (execution/model_configuration.json, sha256 a40e96e212ab87da...). Important clarification (preregistered and enforced): temperature = null indicates the client did not set an explicit temperature parameter; null does not imply temperature=0 or provider-side determinism. We enforced shared_initial_candidate semantics: for each pair we obtained exactly one sampled initial candidate (shared) which both baseline and guarded pipelines analyzed. Therefore any between-condition difference can only arise from (i) guarded-stage emulation observations not encoded as a validator rule, or (ii) the allowed downstream repair-stage invocation(s) recorded in run accounting.

Validator and guarded pipeline implementation. Baseline: deterministic_netops_validator_v5 performing full intent-constraint validation (syntax, command ordering/workflow, required-field presence, and policy-level constraints) and returning binary detected/not-detected per the preregistered scoring rule full_intent_constraint_validation. Guarded: baseline validator followed by an automated sandboxed-emulation harness executing the candidate against an emulated instance of the declared topology and producing runtime observations. The repair policy permitted at most one repair-stage model call per task; run accounting records exactly one repair-stage call in the confirmatory execution.

Execution controls, provenance, and deterministic reconciliation. The adapter family used: hosted_netops_gvr_v1. Execution manifest and deterministic reconciliation artifacts record completed_episode_count = 1440, complete_pair_count = 720, failed_episode_count = 0, and provider-call audit information including model_calls_used = 721 and stage_counts: shared_initial_generation = 720; repair = 1. Provenance and contamination controls (deterministic hashing, artifact-version checks) were applied per the preregistration; no contamination flags occurred for the confirmatory set.

IV. RESULTS

Primary confirmatory counts and test statistics (artifact-grounded). Analysis artifacts report the paired contingency counts: n11 = 719 (detected by both), n10 = 1 (guarded-only), n01 = 0 (baseline-only), n00 = 0 (neither). Marginal detection rates are baseline = 719/720 = 0.9986111111111111 and guarded = 720/720 = 1.0. The paired estimate (guarded - baseline) = 0.0013888888888888884. Exact McNemar two-sided test p = 1.0. The paired-bootstrap percentile 95% CI (10000 resamples, bootstrap_seed = 1729) = [0.0, 0.004166666666666667]. These numbers are reproduced verbatim from analysis/condition_summary.csv (sha256 394bc690671b731194f9b42b4dcbe28fded6e2ec8cf898b79831225343c22c28) and analysis/paired_contingency_table.csv (sha256 efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c).

Canonical artifact pointers and the unique discordant episode. The complete per-episode raw results are archived in execution/raw_results.jsonl (input-results SHA-256 = df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8). The single discordant pair (the unique n10 item) is the lone entry in execution/raw_results.jsonl whose recorded baseline_score == 0 and guarded_score == 1. That raw_results entry contains the canonical pair_id field and includes the validator_trace and repair_provider_trace fields that link the guarded detection to a downstream repair-stage provider trace. Deterministic reconciliation and provider-call accounting are archived in analysis/deterministic_reconciliation.json (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa) and confirm model_calls_used = 721 and stage_counts = {"shared_initial_generation": 720, "repair":

1}. In short: the discordant episode is uniquely identifiable within the archived bundle (execution/raw_results.jsonl, sha256 df367ecc...) and the same bundle contains the deterministic_reconciliation artifact (sha256 49731a2e5...) that documents the single repair-stage call that is connected to that raw_results entry via its repair_provider_trace field.

Run-accounting diagnostics and sensitivity checks. Provider-call audit (deterministic_reconciliation.json) reports hosted_model_call_event_count = 721, completed_hosted_model_call_event_count = 721, failed_hosted_model_call_event_count = 0, cache_hit_count = 0, cache_miss_count = 721. The preregistered missingness policy and sensitivity analyses were applied; no irrecoverable parser or emulation failures occurred (analysis/missingness_summary.csv documents complete_pairs = 720 and zeros for all failure categories). The primary analysis used the preregistered complete_pair_primary treatment; the preplanned sensitivity (treating irrecoverable technical failures as nondetections) is moot here because there were no such failures in the confirmatory run.

Representative per-episode diagnostics. Representative scoring traces (execution/scoring/task-000001-*.json ... task-000006-*.json) show parsed_command_count values of 4, 2, 6, 4, 3, and 6 respectively and validator_violation_count = 0 for these examples; the validator_version reported in these traces is "5.0". These representative artifacts illustrate that the shared_initial candidates in the confirmatory bank were typically complete with respect to the required command sequences and that validator checks reported validity for the majority of cases. The representative traces are archived in the execution bundle and are consistent with the high baseline success rate.

Attribution of the single discordant detection. Because shared_initial_candidate semantics were enforced (one shared sampled initial candidate per pair), the only proximate causes for the single n10 observation are (A) an emulation observation not encoded as a validator rule that produced a guarded-stage detection, or (B) the single recorded repair-stage invocation documented in provider-call accounting. The raw_results entry for the discordant pair includes the repair_provider_trace and validator_trace fields permitting an auditor to follow the exact artifact-level linkage from the guarded detection to the recorded repair-stage call; both the raw_results file (sha256 df367ecc...) and the reconciliation artifact (sha256 49731a2e5...) are present in the canonical execution bundle.

Implications of the quantitative diagnostics. The observed discordant count (n10 = 1) yields an aggregate paired difference that is two orders of magnitude below our preregistered minimum effect-of-interest (0.20). The bootstrap CI [0.0, 0.004166666666666667] quantifies the narrowness of the realized uncertainty in this curated regime. Combined with run-accounting (model_calls_used = 721; repair = 1) and the representative per-episode diagnostics, the archived evidence supports a ceiling-effect interpretation: the deterministic validator covered the task bank's explicit syntactic and workflow constraints so thoroughly that the guarded-stage emulation had

almost no additional aggregate detections to surface.

V. DISCUSSION

Why was the guarded-stage aggregate benefit negligible? The archived evidence supports three artifact-grounded factors.

1) Validator ceiling effect. Deterministic_netops_validator_v5 covered the syntactic and explicit semantic properties encoded in the synthetic task bank (shutdown-configure-restore, make-before-break uplink switchover, paired MTU changes). Baseline success rate ≈ 0.9986 indicates near-universal validator coverage for these curated patterns, leaving negligible headroom for the guarded stage.

2) Task curation and inclusion policy. Confirmatory items were parser-validated synthetic tasks specifically designed to exercise workflow policies but not constructed adversarially to evade validator checks. This curation concentrates cases inside the validator's coverage envelope and reduces the prevalence of subtle runtime-only failure modes that emulation might reveal.

3) Shared_initial_candidate semantics and limited repair activity. Enforcing one sampled initial candidate per pair isolates guarded-stage contributions but removes between-condition generation variability. Provider_call_audit and deterministic_reconciliation.json record one repair-stage invocation aligned with the single n10 discordant count; thus the observed discordance is attributable to downstream guarded-stage activity recorded in run accounting rather than generation differences.

Operational implications and boundary conditions. For environments where a high-coverage static validator exists and inputs match its expected patterns (parser-validated, constrained intents), adding automated sandboxed emulation may provide negligible aggregate additional detection at scale. This conditional conclusion must not be generalized: in operational regimes with noisier natural-language intents, adversarial inputs, broader vendor diversity, stateful multi-step interactions, or validator gaps, guarded runtime checks and higher-fidelity emulation are likely to provide substantive benefit. The archived per-episode traces support targeted follow-up hypotheses: adversarial task injection, broader vendor-CLI coverage, and complementary validator families should increase the guarded-stage marginal contribution.

VI. LIMITATIONS

- Synthetic task bank and deterministic task generator produce limited ecological diversity and create a validator-ceiling effect; results may not generalize to noisier, adversarial, or production distributions.
- Single declared/requested model family (gpt-5-mini) and single validator implementation (deterministic_netops_validator_v5) limit generalizability across models and validator families.
- Preregistered shared_initial_candidate semantics (one initial generation reused across paired conditions) isolate guarded-stage contribution but remove between-condition

generation variability; this design choice constrains discordance sources to downstream guarded-stage observations or repair calls.

- Sandboxed-emulation fidelity relative to vendor/OS production behaviors and large-scale stateful interactions is limited by the emulation harness used; higher-fidelity emulators could change outcomes in other regimes.
- Study power planning targeted a minimum detectable paired difference of 0.20; the observed aggregate effect (~ 0.0014) lies far below that design sensitivity and should be interpreted as a narrowly scoped near-zero finding for this experimental regime rather than a general negative result for all NetOps workflows.
- Provider-side nondeterminism and hidden caching remain residual uncertainties despite deterministic reconciliation procedures; these are documented in `analysis/deterministic_reconciliation.json` and `provider_call_audit` artifacts.

VII. CONCLUSION

We preregistered and executed a paired confirmatory study comparing a strong deterministic validator with the same validator augmented by sandboxed emulation on a curated synthetic intent→configuration task bank. Over 720 paired tasks the guarded pipeline produced a negligible aggregate detection gain (0.0013888888888888884) with exact McNemar $p = 1.0$ and 95% bootstrap CI [0.0, 0.004166666666666667]. Run accounting shows one repair-stage invocation corresponding to the single guarded-only detection; because `shared_initial_candidate` semantics were enforced, archived per-episode traces permit independent verification of that mapping via `execution/raw_results.jsonl` and `analysis/deterministic_reconciliation.json`. We interpret the result as arising from validator ceiling effects and curated task design; follow-up work should focus on adversarial and production-like task banks, multi-validator ensembles, and higher-fidelity emulation to identify regimes where guarded stages add substantive operational value.

Reproducibility pointers (compact). The canonical execution bundle archived by the run contains: `execution/execution_manifest.json` (execution summary and preregistration SHA-256), `execution/raw_results.jsonl` (input-results SHA-256 `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`), `analysis/deterministic_reconciliation.json` (sha256 `49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa`), `execution/model_configuration.json` (sha256 `a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820`), `analysis/paired_contingency_table.csv` (sha256 `efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c`), and `analysis/condition_summary.csv` (sha256 `394bc690671b731194f9b42b4dcbe28fdd6e2ec8cf898b79831225343c22c28`).

The discordant episode is the unique `raw_results.jsonl` entry with `baseline_score = 0` and `guarded_score = 1`; that record contains the `pair_id` and validator/emulation traces that

map it to the single repair-stage invocation recorded in `deterministic_reconciliation.json`. These artifacts are archived as the canonical reproducibility bundle referenced by the execution manifest and the preregistration record. (See Disclosure for exact manifest references.)

DISCLOSURE STATEMENT

Disclosure: Autonomous post-lock research run executed under the frozen master prompt supplied to the system. Declared/requested model: "gpt-5-mini" (provider recorded as `openai_responses`). Deterministic validator: `deterministic_netops_validator_v5`. Orchestration adapter family: `hosted_netops_gvr_v1`. Preregistration id: `f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e` (preregistration SHA-256: `0169919fb8c5aedb2c5a78e031f1ab5c3aeda405cde1ae80fb2f864768595c1b`; recorded in the execution manifest). Initial master-prompt immutable reference: `provenance/master_prompt.txt`, SHA-256 = `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba`. Execution summary (artifact-grounded): `completed_episode_count = 1440`, `complete_pair_count = 720`, `model_calls_used = 721` (720 shared initial + 1 repair-stage call); `stage_counts` and `provider-call audit` are archived in `analysis/deterministic_reconciliation.json` (sha256 `49731a2e52e0...`). Human role and autonomy boundary: a human Research Director selected the broad NetOps topic family and constrained the pre-lock master prompt; after the autonomy lock the agentic pipeline autonomously performed literature selection, preregistration, code and prompt generation, experiment execution, analysis, peer review, and manuscript drafting. No manual human editing of technical manuscript content occurred after the autonomy lock. The execution manifest and archived artifacts named above serve as the canonical reproducibility bundle.

REFERENCES

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- [2] Matt Brown, Ari Fogel, Daniel Halperin, Victor Heorhiadi, Ratul Mahajan, Todd Millstein, "Lessons from the evolution of the Batfish configuration analysis tool", 2023, 10.1145/3603269.3604866
- [3] <http://arxiv.org/abs/2407.08249>